COSC3106 Python Programming Studio
Semester 2 Flex - 2025
'Investigating Preventable Infectious Diseases'
Studio Project Requirements Document

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1. Overview

This document outlines the tasks for the "Preventable Infectious Diseases" social challenge. It serves as both the **product idea** from the User-Centered Design (UCD) perspective and the **case study** from the Database Design perspective.

1.1. Challenge: Investigating Vaccination Data

Vaccination has been a pivotal method in eradicating diseases like smallpox and achieving global herd immunity for others, such as polio. The World Health Organization (WHO) gathers data from member countries to assess vaccination rates, incident rates, and mortality rates for preventable diseases. Our challenge is to develop a web-application that allows various stakeholders to explore detailed and unbiased vaccination data.
You will need to:

- Present this information in an informative, respectful, and unbiased manner.
- Cater for a diverse range of users who are seeking to become more informed on this topic.
- Provide diverse types of information on functionality, including both high-level summaries and enabling an in-depth analysis of the data.

By using your website, users should be able to become well-informed about global immunisation efforts and their impacts.

1.2. Notes on the provided database

You have been provided with a database of immunisation data for the project. You will need to explore the data available, the format, any incomplete or inconsistent data, and the structure of the database.

1.3. Notes on examples provided in this specification

Sample data in this report is for illustration purposes only and does not reflect the actual data or figures in your dataset. You should use these examples as guides whilst you develop a set of features for your identified target groups. There must be a clear distinction between the tasks you implement in sub-tasks A & B.

2. Project Level Details (Web Pages)

This section details the functional requirements of the web-app, organised into 6 pages. Each student will create three pages. Each page corresponds to a sub-task within a level. Each level corresponds an increasing level of complexity. E.g. Student A will complete Sub-Task A tasks of levels 1, 2 and 3. Student B will complete Sub-Task B tasks of levels 1, 2 and 3.

2.1 Level 1 - “Big Picture” Content

Level 1 sub-tasks capture the attention of the users and show how to use your website. It is up to you to decide the best way to show the users the required information, that is, complete the UCD.

Sub Task A: Landing Page

- Capture the attention of all users of your website.

- Highlight the topics targeted by your website.
- Present a snapshot of the data covered by your website.
  You should present 4 facts that are of interest to your selected user groups.

  For example:
  - Display the timeframe covered by the database, e.g., "Vaccination data available from 2000 to 2025."
  - Total number of Vaccines administered
  - Total number of cases of preventable diseases
  - List all the infectious diseases

Sub-Task B: Mission Statement

This presents the overall purpose of your website. This should:

- Present your perspective on how your website addresses the social-challenge.
- Describe how your site can be used.
- Present the Personas that your website targets. **These must be stored in and retrieved from your database.**
- List the names and student numbers of all team members. **These must be stored in and retrieved from your database.**

2.2 Level 2 - "Shallow Glance" of the data

Level 2 sub-tasks explore the issue of vaccination and health data across different dimensions, including geographic, economic, and temporal data. You should carefully consider your UX/UI designs and the database. The sub-tasks include UX/UI design and having a strong understanding of the database structure. In this level, you must demonstrate the ability to select, filter, sort, join and aggregate data using SQL queries efficiently.
Any data anomalies such as missing data, duplicate data etc must be identified and handled appropriately. For example, you might average values for duplicate rows that have inconsistent data.

Sub-Task A: Focused view of vaccination rates by country and region

Present information about the vaccination rates.

- Enable users to filter and organise data by specific countries or regions.
- Allow the user to select a country and/or region
- Allow the user to select a year
- Summarise data to show vaccination rates and herd immunity levels per region

For example, you might display two tables for your user:

- **Table 1: Present information about all countries that have met at least 90% of their vaccination targets for a specific antigen in a specific year.**

Antigen	Year	Country	Region	Percentage of target
MCV2	2010	Switzerland	Western Europe	100

MCV2	2010	Belgium	Western Europe	95
MCV2	2010	Canada	North America	92

- **Table 2: For each region show how many of its countries met at least 90% of their vaccination targets for a specific antigen in a specific year.**

Antigen	Year	Countries that met 90% of target	Region
MCV2	2010	5	South America
MCV2	2010	8	North America
MCV2	2010	2	Australia
MCV2	2010	5	Africa
MCV2	2010	11	Western Europe
MCV2	2010	3	South Asia

Sub-Task B: Focused view of infection data by economic status

Present information about infections of a specific type across all countries belonging to a specific economic status

- Allow the user to select any ONE from all the available economic statuses
- Allow the user to select an infection type
- Allow the user to select a year
- Enable users to filter and organise the data on any of the resultant columns

  For example: Show the country name and measles cases per 100,000 people for all countries with an economic phase of developing in 2022

Preventable disease	Country	Economic phase	Year	Cases per 100,000 people
MEASLES_CASES	Kazakhstan	Developing	2022	7.20
MEASLES_CASES	Tanzania	Developing	2022	9.01

- Allow the user to summarise the data by combining information from more than one table.

  For example: List the total number of cases of rubella for each economic phase in the year 2010

Preventable disease	Economic phase	Year	Cases
Rubella	Developed	2010	9
Rubella	Developing	2010	1807
Rubella	Economy in Transition	2010	1500
Rubella	Least Developed	2010	3987

2.3 Level 3 - “Deep-dive” of the data

Level 3 sub-tasks explore a deeper view of worldwide vaccination and infection data.
In this level, you must demonstrate the ability to use the results of one query to find another sub-dataset. You should carefully consider your UX/UI designs and the ER model. These sub-tasks include complex UX/UI design and ERM design challenges. For top-marks, you will need to take a very well-considered approach.

Write an efficient SQL query that:

- Retrieves only the required data.
- Uses JOINs and column selection to work in a single query if possible.
- Performs sorting based on similarity using SQL.
- Avoids post-processing of data using Python, where it can be handled through careful crafting of the SQL query.
- Sorts results by a user-selected criterion.
- Note that this is a challenge in making suitable SQL queries.

Sub-Task A: Identify countries with the biggest improvement in vaccination rates

In this task, enable users to discover which countries have the highest jump in vaccination rates from a period of time—denoted with a starting and ending year—for a specific antigen type.
Note: the vaccination rate needs to be calculated and depends on the population of the country
Your user should be able to:

- Select a starting year and an ending year
- Choose an antigen type
- Choose the number of countries to find

For Example:
The user wants to find the top ten countries that have the largest vaccination rate jump from 2020 to 2024.

Country	Vaccination Rate Increase	Start Year	End Year
Uganda	5.6%	2000	2024
Mozambique	4.2%	2000	2024
Uzbekistan	2.3%	2000	2024

Sub-Task B: Identify countries with above average infection rate

In this task, enable your users to find the global reported infection per 100,000 people and the countries whose reported infection per 100,000 people exceeds that global average for a specific year and infection type

Your user should be able to:

- Select a year
- Select an infection type
- View both the global infection rate and the data of countries whose infection rate exceeds the global rate

- The global infection rate should be shown at the top of the table for easy viewing.

For Example:

The user wants to find the global measles infection rate per 100,000 people and the countries whose rate exceeds that in the year 2020

Country	**Infection Type**	**Infection per 100,000 people**	**Year**
Global	Measles	1.03	2020
Afghanistan	Measles	1.64	2020
Brazil	Measles	4.05	2020
Ethiopia	Measles	3.65	2020
Kazakhstan	Measles	4.02	2020